Case Records of the Wexford General Hospital

A Clinicopathological Exercise • Synthetic Teaching Case

Case 28-2026: A 55-Year-Old Man with Chronic Diarrhea, Weight Loss, and Migratory Arthralgia

The following is an entirely fictional teaching case. All names, dates, case numbers, institutions, laboratory values, and narrative details are invented for instructional purposes and do not depict any real patient, physician, or institution.

Presentation of the Case

Dr. Marcus Ellery (Medicine): A 55-year-old man was admitted to this hospital because of nine months of chronic diarrhea and progressive weight loss, on a background of several years of migratory joint pain.

The patient had been well until approximately four years before the current admission, when he began to have episodes of pain and swelling in the large joints. The arthritis was migratory and asymmetric, affecting first the knees, then the wrists, and later the ankles and shoulders, with each episode lasting days to a few weeks before resolving and reappearing elsewhere. The joints were at times frankly swollen and warm but never became deformed, and there was no morning stiffness of prolonged duration. He was evaluated by a rheumatologist; serologic testing for rheumatoid factor, anti–cyclic citrullinated peptide antibody, and antinuclear antibody was reportedly negative on more than one occasion, and radiographs showed no erosions. A diagnosis of seronegative inflammatory arthritis was made. He was treated initially with nonsteroidal antiinflammatory drugs (NSAIDs), with partial relief, and subsequently with a short course of low-dose prednisone. Because of persistent symptoms, a tumor necrosis factor (TNF) inhibitor, adalimumab, was begun approximately 18 months before admission and continued for several months; the patient recalled that, although his joints improved modestly, his general health seemed to decline during this period, with the onset of fatigue and intermittent loose stools, and the agent was eventually discontinued.

Approximately nine months before admission, the patient began to have diarrhea. The stools were initially loose and watery, occurring three to six times daily, and over the ensuing months became bulky, pale, greasy, foul-smelling, and difficult to flush, which he and his wife attributed to "fat in the stool." The diarrhea was accompanied by postprandial abdominal bloating, crampy periumbilical and lower abdominal pain, and audible borborygmi. There was no hematochezia, melena, or nocturnal awakening to defecate, although the diarrhea did occur on some nights. During the same interval he had progressive, unintentional weight loss that he estimated at 12 kg, anorexia, profound fatigue, and intermittent low-grade fevers with occasional night sweats. His wife had noticed, in the months before admission, that he had become more forgetful, slower in conversation, and less engaged than was his custom, and that he occasionally lost the thread of a task. He had also noted that his skin, particularly on the face, neck, and forearms, had become darker over the preceding year, which he had ascribed to outdoor work. There was mild swelling of both ankles. He denied dysphagia, odynophagia, vomiting, jaundice, cough, dyspnea, chest pain, palpitations, headache, visual change, diplopia, focal weakness, paresthesia, dysuria, or rash.

He was evaluated by his primary physician several months before admission, at which time empiric trials of a lactose-free diet and, later, of loperamide produced no meaningful benefit. A two-week course of metronidazole, given for presumed infectious or parasitic enteritis, likewise produced no improvement. Because of continued weight loss and the development of ankle edema, he was referred for inpatient evaluation.

His medical history was notable only for the seronegative arthritis described above and for an appendectomy in adolescence. He took NSAIDs intermittently for joint pain; he had completed the courses of prednisone, adalimumab, metronidazole, and loperamide noted above and was taking no other medications, herbal products, or proton-pump inhibitors at the time of admission. He had no known drug allergies. His immunizations were up to

date, including seasonal influenza and pneumococcal vaccination.

The patient was a lifelong resident of a rural farming community in the central United States and worked as a farmer; he was extensively exposed to soil, livestock, manure, and water from a private well, and he had participated for years in the maintenance of the farm's septic and wastewater systems. He lived with his wife of three decades and reported a monogamous heterosexual relationship; there was no history of sexually transmitted infection. He had never traveled outside the United States and had not visited the tropics. He had never smoked, drank two to three beers per week, and had never used injection or other illicit drugs. He drank well water and occasionally consumed unpasteurized milk from the farm. There was no known tuberculosis contact and no history of incarceration or homelessness.

His family history included type 2 diabetes mellitus in his father and coronary disease in both parents; there was no family history of inflammatory bowel disease, celiac disease, autoimmune disease, or gastrointestinal cancer.

On review of systems, he confirmed low-grade fevers, night sweats, the skin darkening, the ankle swelling, and the cognitive changes described by his wife. He denied oral ulcers, sicca symptoms, photosensitivity, Raynaud phenomenon, eye pain or redness, back pain of inflammatory character, dyspnea, orthopnea, hematuria, or new neurologic events.

On examination, the patient appeared chronically ill, cachectic, and older than his stated age. The temperature was 37.9°C, the heart rate 96 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 98% while he was breathing ambient air. The blood pressure was 112/70 mm Hg while he was supine and fell to 94/60 mm Hg on standing, with a rise in heart rate to 112 beats per minute and reported light-headedness, indicating orthostatic intolerance. The body-mass index was 19.0. There was diffuse hyperpigmentation that was most pronounced over sun-exposed areas — the face, the posterior neck, and the dorsal forearms — but that also involved skin creases; the buccal mucosa was not pigmented. There was no scleral icterus and no conjunctival pallor of note. Small, soft, mobile, nontender lymph nodes were palpable in the cervical, axillary, and inguinal regions, none exceeding 1.5 cm. The jugular venous pressure was not elevated. Cardiac auscultation revealed a regular rhythm with a soft (grade 2/6) early diastolic murmur best heard at the left sternal border; there were no peripheral stigmata of endocarditis. The lungs were clear. The abdomen was mildly distended with active bowel sounds; there was mild, diffuse tenderness without guarding or rebound, no palpable mass, and shifting dullness suggestive of a small amount of ascites; the liver edge was palpable 2 cm below the right costal margin and the spleen tip was just palpable. There was pitting edema to the mid-shins bilaterally. On musculoskeletal examination there was no active synovitis, effusion, or deformity in any joint at the time of presentation. On neurologic examination he was alert and fully oriented but exhibited mild psychomotor slowing; bedside cognitive testing showed reduced verbal fluency and difficulty with serial calculation and abstraction, with a pattern suggesting frontal-executive dysfunction, while naming and recall were relatively preserved. Cranial-nerve examination, including a careful assessment of extraocular movements, was normal; specifically, there was no supranuclear gaze palsy and no rhythmic, pendular convergence of the eyes synchronized with contraction of the jaw muscles — that is, no oculomasticatory myorhythmia. Strength, tone, reflexes, coordination, and sensation were normal, and there was no meningismus.

Laboratory studies obtained on admission are shown in **Table 1**, **Table 2**, and **Table 3**.

Table 1. Hematologic and Coagulation Studies.

Variable	Patient's Value	Reference Range
Hemoglobin (g/dL)	10.2	13.5–17.5
Hematocrit (%)	31.4	41.0–53.0
Mean corpuscular volume (fL)	84	80–100
Red-cell distribution width (%)	17.8	11.5–14.5

Variable	Patient's Value	Reference Range
White-cell count (×10⁹/L)	8.9	4.5–11.0
Neutrophils (%)	72	40–70
Lymphocytes (%)	18	22–44
Platelet count (×10⁹/L)	498	150–400
Peripheral smear	Dimorphic red cells with microcytic, hypochromic forms and occasional macro-ovalocytes; mild anisopoikilocytosis; no schistocytes; no blasts	—
Prothrombin time (s)	16.8	11.5–14.5
International normalized ratio	1.4	0.9–1.1
Activated partial-thromboplastin time (s)	33	25–35

Table 2. Clinical Chemistry and Nutritional Studies.

Variable	Patient's Value	Reference Range
Sodium (mmol/L)	134	135–145
Potassium (mmol/L)	3.2	3.5–5.0
Chloride (mmol/L)	108	98–107
Carbon dioxide (mmol/L)	20	22–30
Urea nitrogen (mg/dL)	24	8–20
Creatinine (mg/dL)	1.1	0.7–1.3
Glucose (mg/dL)	88	70–99
Calcium (mg/dL)	7.9	8.5–10.5
Magnesium (mg/dL)	1.4	1.7–2.4
Phosphorus (mg/dL)	2.3	2.5–4.5
Total protein (g/dL)	5.2	6.0–8.3
Albumin (g/dL)	2.5	3.5–5.0
Aspartate aminotransferase (U/L)	31	10–40
Alanine aminotransferase (U/L)	28	7–56
Alkaline phosphatase (U/L)	96	40–130
Total bilirubin (mg/dL)	0.6	0.2–1.2
Iron (µg/dL)	28	65–175
Total iron-binding capacity (µg/dL)	198	250–450
Transferrin saturation (%)	14	20–50
Ferritin (ng/mL)	14	30–400
Vitamin B12 (pg/mL)	142	200–900
Folate (ng/mL)	2.1	>4.0
25-Hydroxyvitamin D (ng/mL)	11	30–80
Thyrotropin (µIU/mL)	1.8	0.4–4.0

Table 3. Inflammatory, Stool, and Serologic Studies.

Variable	Patient's Value	Reference Range
Erythrocyte sedimentation rate (mm/hr)	70	0–20

Variable	Patient's Value	Reference Range
C-reactive protein (mg/L)	85	<5
Quantitative fecal fat (g/24 hr)	28	<7
Fecal calprotectin (µg/g)	165	<50
Fecal elastase-1 (µg/g)	410	>200
Stool ova and parasites (×3)	Negative	Negative
Stool *Giardia* antigen	Negative	Negative
Stool culture and *Clostridioides difficile* toxin	Negative	Negative
Tissue transglutaminase IgA antibody (U/mL)	<2	<4
Total IgA (mg/dL)	245	70–400
HIV antigen/antibody (4th-generation)	Negative	Negative
Tuberculosis interferon-γ release assay	Negative	Negative
Serum protein electrophoresis	No monoclonal protein; polyclonal hypogammaglobulinemia not present	—

Three sets of blood cultures obtained over 48 hours showed no growth at five days. A urinalysis was unremarkable. Antinuclear antibody, antineutrophil cytoplasmic antibodies, rheumatoid factor, and anti–cyclic citrullinated peptide antibody were again negative, and serum complement levels were normal.

Dr. Priya Raman (Radiology): Computed tomography (CT) of the abdomen and pelvis, performed after the administration of intravenous and oral contrast material, revealed diffuse, marked thickening of the folds of the proximal small bowel, most pronounced in the duodenum and jejunum, with a somewhat feathery, edematous appearance of the valvulae conniventes. There was bulky lymphadenopathy in the small-bowel mesentery and the retroperitoneum, with several nodes measuring 2 to 4 cm in short axis. A striking feature was that many of these enlarged nodes were of low attenuation, with central density approaching that of fat — a finding attributable to the accumulation of lipid and to the architectural effacement that characterizes certain infiltrative and infectious adenopathies, rather than to the soft-tissue density expected of most malignant or reactive nodes. There was mild hepatomegaly and splenomegaly and a small volume of ascites. No discrete bowel mass, no bowel-wall stratification suggestive of transmural Crohn inflammation, no fistula or abscess, no biliary or pancreatic duct dilatation, and no focal hepatic or splenic lesion was identified. The pancreas appeared normal in size and attenuation.

Dr. Theodore Onyeka (Gastroenterology): Upper endoscopy was performed. The esophagus and stomach were normal. The mucosa of the second and third portions of the duodenum was abnormal throughout: it appeared pale and edematous, with a shaggy, granular surface studded with small yellowish-white plaques and white-tipped villi, and there were scattered areas of friability that bled lightly on contact. The mucosal folds were thickened, and there was a diffuse, almost frosted appearance consistent with dilated, lipid-laden lymphatics (lymphangiectasia). Multiple biopsy specimens were obtained from the distal duodenum and proximal jejunum. Given the constellation of findings, additional specimens were sent for special staining and for molecular testing, and the neurology service was consulted regarding cerebrospinal fluid sampling.

Dr. Marcus Ellery (Medicine): A diagnostic procedure was performed.

Differential Diagnosis

Dr. Adaeze Okoro (Infectious Diseases): The diagnostic problem is best approached through the chronology, which is the most important clue in the case.

This is a previously healthy 55-year-old farmer with a multisystem illness that has evolved over roughly four years in two distinct phases. The first phase, lasting some three years, was a *seronegative, migratory, large-joint inflammatory arthritis* — knees, wrists, ankles,

shoulders — that was nondestructive, was repeatedly negative for rheumatoid factor, anti–cyclic citrullinated peptide antibody, and antinuclear antibody, and that responded only partially to NSAIDs, prednisone, and a TNF inhibitor. The second phase, beginning about nine months before admission and overlapping with the tail of his immunosuppressive treatment, is a *malabsorptive gastrointestinal illness*: chronic, ultimately steatorrheal diarrhea, postprandial bloating and cramping, and 12 kg of weight loss, accompanied by *systemic and extraintestinal features* — low-grade fever and night sweats, generalized lymphadenopathy, hyperpigmentation, hypoalbuminemic edema, a soft early diastolic murmur, and subtle frontal-executive cognitive decline. At least five domains require unification: chronic arthropathy, malabsorption, constitutional and febrile symptoms, lymphadenopathy with splenomegaly, and central-nervous-system and possible cardiac involvement.

The laboratory data are entirely consistent with a small-intestinal malabsorptive process superimposed on chronic systemic inflammation. He has a dimorphic anemia with both iron deficiency (low iron, low transferrin saturation, low total iron-binding capacity, low ferritin) and deficiencies of folate and vitamin B12, with a wide red-cell distribution width and a dimorphic smear — a pattern expected when the proximal small bowel, where iron and folate are absorbed, and the terminal ileum, where B12 is absorbed, are both diseased. The prolonged prothrombin time with a normal aPTT reflects vitamin K deficiency from fat malabsorption; the very low 25-hydroxyvitamin D, the hypocalcemia, and the hypomagnesemia tell the same story. The hypoalbuminemia and low total protein, together with the edema and ascites, indicate protein loss and undernutrition — a protein-losing enteropathy from the dilated lymphatics is very likely. The elevated quantitative fecal fat confirms steatorrhea objectively. The markedly elevated erythrocyte sedimentation rate and C-reactive protein confirm that this is an inflammatory, not merely a nutritional, disorder. Several common explanations have already been *excluded by the workup*: the fecal elastase is normal, which makes exocrine pancreatic insufficiency very unlikely; the thyrotropin is normal, excluding hyperthyroidism as a cause of weight loss and diarrhea; the stool studies, including three ova-and-parasite examinations and a *Giardia* antigen test, are negative; and the HIV antigen/antibody test is negative. The differential diagnosis of chronic malabsorptive diarrhea with extraintestinal disease can now be considered.

Celiac disease

Celiac disease is the most common cause of chronic malabsorption with proximal small-bowel involvement in an adult of European ancestry, and it can produce iron, folate, and fat-soluble-vitamin deficiencies, weight loss, hypoalbuminemia, and even an associated arthropathy and dermatologic findings. The endoscopic appearance of scalloped, atrophic duodenal folds can superficially resemble what was seen here. However, two findings argue strongly against it. First, the tissue transglutaminase IgA antibody is negative, and — importantly — the total IgA level is normal, so this is not a false negative caused by IgA deficiency. Seronegative celiac disease exists, but it is uncommon. Second, and more tellingly, the endoscopic and CT findings are *not* those of celiac disease: celiac disease causes villous atrophy, not the shaggy, yellowish-white, plaque-studded, lipid-laden mucosa described here, and it does not produce bulky, low-attenuation mesenteric and retroperitoneal lymphadenopathy. The years of preceding migratory arthritis, the fever, the lymphadenopathy, and the cognitive change are all foreign to uncomplicated celiac disease. Refractory celiac disease can be complicated by lymphoma, but celiac disease itself is unlikely.

Crohn disease and other inflammatory bowel disease

Crohn disease can cause chronic diarrhea, weight loss, abdominal pain, malabsorption (particularly with proximal or extensive small-bowel involvement), fever, arthritis, and a raised calprotectin, and the migratory large-joint arthropathy of inflammatory bowel disease is well described. However, the diarrhea here is steatorrheal and non-bloody, the CT shows diffuse fold thickening rather than the segmental, transmural, stratified bowel-wall inflammation, skip lesions, strictures, fistulas, or abscesses that characterize Crohn disease, and the bulky low-attenuation adenopathy is atypical. The fecal calprotectin is only mildly elevated, which is more in keeping with a low-grade or infiltrative mucosal process than with active transmural Crohn disease. Most importantly, Crohn disease would not explain the hyperpigmentation, the generalized peripheral lymphadenopathy, the diastolic murmur, or the cognitive decline. It does not unify the case.

Intestinal and other lymphoma

This is the most important "cannot-miss" alternative. A small-bowel lymphoma — particularly enteropathy-associated T-cell lymphoma, but also other peripheral T-cell lymphomas, mantle-cell lymphoma with lymphomatous polyposis, or an immunoproliferative small-intestinal disease (alpha–heavy-chain disease) — can produce chronic diarrhea, malabsorption, weight loss, fever, night sweats, hepatosplenomegaly, and bulky mesenteric and retroperitoneal lymphadenopathy. The history of a TNF inhibitor is relevant, because immunosuppression raises lymphoma risk, and B symptoms with adenopathy and splenomegaly always demand that lymphoma be excluded histologically. Several features, however, sit uneasily with lymphoma. The lymphadenopathy is of strikingly *low, near-fat attenuation*, which is unusual for untreated nodal lymphoma; the nodes are generalized and small peripherally; and the duodenal mucosa shows a diffuse, infiltrative, lipid-laden, plaque-like change rather than a mass, ulcer, or polypoid lesion. Above all, lymphoma does not readily explain a *three-year antecedent migratory arthritis*. Lymphoma cannot be excluded on clinical grounds — only the biopsy can do that — but it is not the most parsimonious explanation, and the biopsy should be examined specifically for it.

Chronic infections of the small bowel

Several chronic infections deserve attention. *Mycobacterium avium* complex (MAC) infection of the bowel is the classic infectious mimic of Whipple disease: it fills the lamina propria with macrophages that are periodic acid–Schiff (PAS)–positive and produces malabsorption, fever, weight loss, lymphadenopathy, and hepatosplenomegaly. But disseminated MAC enteritis occurs almost exclusively in profoundly immunosuppressed patients, especially those with advanced HIV infection and very low CD4 counts — and HIV-negative patients are uncommonly affected. That single negative result substantially reduces the probability of MAC and sets up the crucial histochemical distinction in this case. *Intestinal tuberculosis* can cause chronic diarrhea, weight loss, fever, and mesenteric adenopathy — and tuberculous nodes can show low-attenuation, necrotic centers on CT — but it characteristically involves the ileocecal region with ulceration and stricture rather than diffuse duodeno-jejunal plaque formation, it would be unusual without pulmonary or other findings, and the interferon-γ release assay is negative. *Giardiasis* is a classic cause of malabsorptive diarrhea acquired from well water and farm exposure, which this man has in abundance; but the stool *Giardia* antigen and three ova-and-parasite examinations are negative, and a course of metronidazole produced no benefit, making giardiasis very unlikely. *Tropical sprue* causes malabsorption with folate and B12 deficiency but requires residence in or travel to an endemic tropical region, which he denies. *Small-intestinal bacterial overgrowth* can produce bloating, diarrhea, steatorrhea, and B12 deficiency, and could even be a secondary phenomenon here, but it does not by itself cause fever, lymphadenopathy, hyperpigmentation, arthritis, or cognitive decline, and it would not explain the endoscopic appearance. The chronic infection that can unify the case is *Whipple disease*, caused by the actinomycete *Tropheryma whipplei*.

Autoimmune and infiltrative enteropathies

Autoimmune enteropathy is a rare cause of refractory malabsorptive diarrhea, typically with circulating anti-enterocyte or anti-goblet-cell antibodies and an association with other autoimmune phenomena; it is principally a diagnosis of childhood but occurs in adults. It would not, however, explain the fever, the adenopathy, the murmur, or the antecedent arthritis, and the histology is distinctive. *Systemic amyloidosis* (AL or AA type) can infiltrate the bowel to cause malabsorption, diarrhea, weight loss, protein-losing enteropathy, hepatosplenomegaly, macroglossia, an infiltrative cardiomyopathy, and autonomic dysfunction — and the orthostatic hypotension here might raise that possibility. AA amyloidosis could even be a downstream consequence of his years of chronic inflammatory arthritis. However, the serum protein electrophoresis shows no monoclonal protein, the diastolic murmur and preserved wall findings are not typical of amyloid heart disease, and amyloid would not produce the specific endoscopic plaques or the low-attenuation adenopathy. It remains a consideration that the biopsy, with a Congo red stain, can readily address.

Endocrine, pancreatic, and paraneoplastic causes

For completeness: *hyperthyroidism* can cause weight loss and hyperdefecation, but the thyrotropin is normal. *Exocrine pancreatic insufficiency* causes steatorrhea, but the fecal elastase is normal and there is no pancreatic abnormality on CT or history of pancreatitis or alcohol excess sufficient to explain it. *Addison disease* deserves

consideration, because it can explain hyperpigmentation, hyponatremia and hyperkalemia, postural hypotension, weight loss, and gastrointestinal symptoms; however, his potassium is *low* rather than high (consistent with diarrheal losses), and adrenal insufficiency does not cause steatorrhea, lymphadenopathy, arthritis, or the described enteropathy. A morning cortisol would be reasonable, since chronic illness and the hyperpigmentation make coexisting adrenal involvement worth excluding. A *paraneoplastic process* or a hormone-secreting neuroendocrine tumor (for example, a VIPoma or gastrinoma) can cause secretory diarrhea, but these produce watery, not steatorrheal, diarrhea and would not explain the multisystem picture or the histology.

Synthesis and convergence

The pieces can now be reassembled. This is a middle-aged white man, a farmer with intense and chronic exposure to soil, livestock, manure, and wastewater, who has, in sequence, a *years-long seronegative migratory nondestructive large-joint arthritis*, followed by a *malabsorptive, steatorrheal, protein-losing enteropathy* with weight loss, *low-grade fever and night sweats, generalized lymphadenopathy, hepatosplenomegaly, hyperpigmentation, a diastolic heart murmur, and subtle frontal cognitive decline*. The endoscopy shows pale, shaggy, yellow-white, lipid-laden duodeno-jejunal mucosa, and the CT shows diffuse jejunal fold thickening with bulky, low-attenuation (approaching fat-density) mesenteric and retroperitoneal lymphadenopathy. Common diagnoses do not unify this constellation. Whipple disease does.

The chronology is the decisive clue. *A seronegative, migratory, large-joint arthritis that precedes a malabsorptive illness by several years is one of the most characteristic presentations of Whipple disease*. In classic Whipple disease, articular symptoms antedate the gastrointestinal phase by a mean of six to eight years; the arthritis is migratory, nondeforming, and seronegative, and it is very frequently misdiagnosed as seronegative spondyloarthropathy or rheumatoid arthritis and treated — as here — with immunosuppression. This point is not merely diagnostic but pathophysiologic: corticosteroids and, especially, TNF inhibitors are well documented to *unmask or accelerate* Whipple disease, and the temporal association between this man's adalimumab therapy and the explosive onset of his gastrointestinal and constitutional decline is exactly the natural history one would predict. *Tropheryma whipplei* is an environmental organism found in soil and sewage, with a recognized excess among farmers, sewage and wastewater workers, and others with occupational soil exposure — his occupation fits the epidemiology precisely. The hyperpigmentation, generalized lymphadenopathy, hepatosplenomegaly, hypoalbuminemic edema (from protein-losing enteropathy through the obstructed, dilated intestinal lymphatics), and chronic low-grade fever are all classic systemic features of the disease. The low-attenuation, near-fat-density adenopathy is a described and relatively specific imaging signature of Whipple disease, reflecting the lipid- and macrophage-laden, architecturally effaced nodes. The diastolic murmur raises the real possibility of *Tropheryma whipplei* culture-negative endocarditis, an increasingly recognized manifestation that must be sought with echocardiography. The cognitive change also compels consideration of central-nervous-system Whipple disease, which carries the gravest prognosis.

The examination is relevant on this point. The clinician specifically looked for, and did not find, *oculomasticatory myorhythmia* — the slow, rhythmic, pendular convergence of the eyes synchronized with contraction of the masticatory muscles. This sign is *pathognomonic* of central-nervous-system Whipple disease, but it is also *rare*, present in only a minority of cases; its absence in no way excludes neurologic involvement. Given his subtle frontal-executive dysfunction and the high stakes of missing neurologic disease, cerebrospinal-fluid analysis with a *Tropheryma whipplei*–specific polymerase-chain-reaction (PCR) assay is mandatory, whether or not the bedside examination is localizing.

The diagnostic test, therefore, is the one that was performed: *esophagogastroduodenoscopy with multiple biopsies of the distal duodenum and proximal jejunum*. The expected histology would show the lamina propria expanded and packed with foamy macrophages containing abundant PAS-positive, diastase-resistant granular material, together with dilated, lipid-filled lymphatics. The single most important confirmatory and discriminating stain is the acid-fast (Ziehl–Neelsen) stain: in Whipple disease the macrophages are PAS-positive but *acid-fast negative*, which distinguishes the disease from *Mycobacterium avium* complex infection, in which the macrophages are PAS-positive but *acid-fast positive*. The diagnosis should be confirmed with a *Tropheryma whipplei*–specific PCR assay on the biopsy tissue, and cerebrospinal fluid should be sent for the same PCR to

assess for subclinical central-nervous-system involvement, because that result determines the intensity and duration of therapy.

Clinical Diagnosis

Chronic malabsorptive (steatorrheal) diarrhea with protein-losing enteropathy, weight loss, multiyear antecedent seronegative migratory large-joint arthritis, generalized lymphadenopathy, hepatosplenomegaly, hyperpigmentation, a diastolic murmur, and subtle cognitive decline — most consistent with systemic Whipple disease (*Tropheryma whipplei* infection) with small-intestinal involvement, unmasked by TNF-inhibitor therapy.

Dr. Adaeze Okoro's Diagnosis

Whipple disease (systemic *Tropheryma whipplei* infection with small-intestinal involvement), with probable central-nervous-system involvement to be confirmed by cerebrospinal-fluid PCR.

Pathological Discussion

Dr. Henrik Solberg (Pathology): The diagnostic procedure was esophagogastroduodenoscopy with multiple mucosal biopsies obtained from the distal duodenum and proximal jejunum. Cerebrospinal fluid was subsequently obtained for molecular testing, and the biopsied mucosal tissue was examined with routine histology, histochemical stains, immunohistochemistry, molecular testing, and electron microscopy.

On hematoxylin-and-eosin–stained sections, the small-intestinal villi were blunted and clubbed, and the lamina propria was markedly expanded and distended by sheets of large, foamy macrophages with abundant pale, granular, eosinophilic-to-amphophilic cytoplasm. Interspersed among the macrophages were conspicuous dilated lymphatic channels and clear, optically empty vacuolar spaces representing extracellular lipid, accounting for the lipid-laden, lymphangiectatic appearance noted at endoscopy and the low attenuation of the involved nodes on imaging. There was relatively little accompanying lymphocytic or neutrophilic infiltrate, and no granulomas, no malignant lymphoid population, and no dysplasia were identified.

The diagnostic stain was the periodic acid–Schiff (PAS) reaction with diastase digestion. The cytoplasm of the macrophages was filled with coarse, abundant, magenta, PAS-positive granules and sickle-shaped inclusions that were *resistant to diastase (amylase) pretreatment*, indicating that the material was not glycogen but the glycoprotein-rich cell walls of accumulated bacilli. This PAS-positive, diastase-resistant macrophage infiltrate of the lamina propria is the histologic hallmark of Whipple disease.

The critical differential at the bench is between Whipple disease and infection by *Mycobacterium avium* complex, because the latter also produces a lamina propria stuffed with PAS-positive macrophages and can appear nearly identical on the PAS stain alone. The distinction is made with an acid-fast stain. A Ziehl–Neelsen stain performed on these sections was *negative*: no acid-fast organisms were identified within the macrophages. In *Mycobacterium avium* complex infection the macrophage inclusions are both PAS-positive and *acid-fast positive*; the combination of PAS-positivity with acid-fast-negativity seen here therefore excludes mycobacterial infection and is characteristic of Whipple disease. This distinction is also why the negative HIV test in this patient was so important, since *Mycobacterium avium* complex enteritis is overwhelmingly a disease of the profoundly immunocompromised.

The diagnosis was confirmed by molecular and immunohistochemical methods. Immunohistochemical staining with a monoclonal antibody directed against *Tropheryma whipplei* strongly and specifically labeled the organisms within the macrophage cytoplasm. A *Tropheryma whipplei*–specific polymerase-chain-reaction assay performed on the biopsy tissue, targeting a segment of the bacterial 16S ribosomal RNA gene and confirmed by sequencing, was positive. Notably, the *Tropheryma whipplei* PCR assay performed on the cerebrospinal fluid was *also positive*, despite the absence of focal neurologic signs and of oculomasticatory myorhythmia, establishing subclinical central-nervous-system involvement — a finding with direct therapeutic consequences.

Finally, transmission electron microscopy of the lamina propria demonstrated, both within macrophage phagolysosomes and free in the extracellular space, abundant small, rod-shaped bacilli measuring approximately 1 to 2 μm in length, each enclosed by a characteristic trilamellar (three-layered) cell wall — the ultrastructural signature of *Tropheryma whipplei*. The degenerating bacterial cell walls within the macrophages correspond to the PAS-positive material seen by light microscopy.

The integrated findings — PAS-positive, diastase-resistant, acid-fast-negative foamy macrophages expanding the small-intestinal lamina propria, with confirmatory *Tropheryma whipplei* immunohistochemistry, positive tissue and cerebrospinal-fluid PCR, and characteristic trilamellar bacilli on electron microscopy — establish the diagnosis of Whipple disease with central-nervous-system involvement.

Discussion of Management

Dr. Beatrice Lund (Infectious Diseases): Whipple disease is a chronic, multisystem bacterial infection that is often fatal if untreated but is curable with prolonged, appropriately chosen antibiotic therapy; the two central principles of management are that treatment must be of long duration and that it must achieve adequate concentrations within the central nervous system, because relapse — particularly cerebral relapse — is the feared outcome and carries a grave prognosis.

This patient's documented cerebrospinal-fluid PCR positivity placed him explicitly in the category of central-nervous-system Whipple disease, and his regimen was constructed accordingly. He received an *induction phase* of intravenous ceftriaxone, 2 g daily for two weeks; a parenteral, central-nervous-system–penetrating beta-lactam (intravenous penicillin G is an accepted alternative) is used for induction precisely to sterilize the central nervous system. This was followed by a prolonged *maintenance phase* of oral trimethoprim–sulfamethoxazole (one double-strength tablet twice daily) for approximately 12 months, the sulfamethoxazole component providing continued central-nervous-system coverage. For teaching purposes, the optimal maintenance regimen is debated: because the organism's intrinsic genomic features and specific enzymatic pathways mediate resistance to the trimethoprim component in many patients, some authorities prefer prolonged oral doxycycline combined with hydroxychloroquine (the latter alkalinizes the phagolysosome and renders doxycycline bactericidal), often for 12 to 18 months, and reserve or combine therapies in central-nervous-system disease. Either approach requires a duration measured in many months, never the short courses that had repeatedly failed this patient in the form of metronidazole and other empiric antibiotics.

Two complications of treatment warranted specific attention. The first is the *immune reconstitution inflammatory syndrome (IRIS)*, which can occur after the initiation of effective antibiotics in Whipple disease, particularly — as in this patient — when the diagnosis was made in the setting of recent immunosuppression. IRIS can manifest as recurrent high fever, arthritis, serositis, or worsening neurologic findings days to weeks into therapy, and it is treated with corticosteroids rather than by abandoning the antibiotic. Because this man had received both prednisone and a TNF inhibitor, which are known to worsen Whipple disease and to predispose to IRIS, he was monitored closely; a transient febrile, arthralgic flare two weeks into therapy was attributed to IRIS and responded promptly to a short course of corticosteroids. The second issue was *nutritional repletion*: he received parenteral and then enteral repletion of iron, folate, vitamin B12, vitamin D, calcium, magnesium, and the fat-soluble vitamins, with vitamin K correcting his prolonged prothrombin time, together with a high-protein diet and attention to his hypoalbuminemia and ascites. Transthoracic and then transesophageal echocardiography were performed because of the diastolic murmur; these showed mild aortic valve thickening without a discrete vegetation, and the murmur was attributed to a low-grade valvulopathy, with the *Tropheryma whipplei* infection regarded as the unifying cause and the prolonged antibiotic course considered adequate to treat possible early endocardial involvement.

The clinical response was excellent. The fevers and night sweats resolved within the first one to two weeks of therapy, the diarrhea and steatorrhea abated over the first month, and the abdominal bloating and cramping resolved. Over the subsequent three to six months he regained approximately 10 kg, the edema and ascites resolved with correction of his hypoalbuminemia, the anemia corrected, and the inflammatory markers normalized. His family reported gradual improvement in his alertness, conversational engagement, and executive function, and repeat cognitive testing showed measurable improvement. Surveillance for relapse — clinical assessment, and repeat duodenal biopsy with PCR and repeat cerebrospinal-fluid PCR at the conclusion of maintenance therapy — was planned, with the understanding that he would require long-term follow-up, because relapse can occur years later and central-nervous-system relapse is frequently refractory.

The teaching points of this case are several. First, *the arthritis comes first*: a seronegative, migratory, nondestructive large-joint arthritis preceding a malab-

sorptive illness by years should raise the possibility of Whipple disease, and the immunosuppression so often prescribed for the presumed rheumatologic disorder can unmask and accelerate the infection — a sequence vividly illustrated here by the TNF inhibitor. Second, *the histochemistry is decisive*: PAS-positive, diastase-resistant macrophages in the small-bowel lamina propria establish the diagnosis, and the acid-fast stain distinguishes Whipple disease (acid-fast negative) from *Mycobacterium avium* complex infection (acid-fast positive), with the patient's HIV status providing essential context. Third, *the central nervous system must always be assessed*, with cerebrospinal-fluid PCR even in the absence of the rare but pathognomonic oculomasticatory myorhythmia, because subclinical neurologic involvement mandates central-nervous-system–penetrating induction therapy and lifelong vigilance for relapse. Finally, the occupational and environmental history — soil, livestock, and wastewater exposure in a farmer — reminds us that *Tropheryma whipplei* is an environmental organism, and that the patient's social history can point toward a diagnosis that is otherwise easily missed.

Anatomical Diagnosis

Whipple disease (systemic *Tropheryma whipplei* infection with small-intestinal involvement), with associated central-nervous-system involvement confirmed by cerebrospinal-fluid polymerase-chain-reaction testing.